PMP CERTIFICATION COURSE

Student Reference Resource Guide

Section 7

Models, Methods & Artifacts

All Lectures Covered

PMBOK Guide 8th Edition | 2026 ECO

Aligned with the PMI Project Management Professional Examination Content Outline

Section Overview

Section 7 of the PMP curriculum, aligned with PMBOK Guide 8th Edition, introduces the formal vocabulary that replaces the older Tools and Techniques chapter found in PMBOK 6 and earlier. PMBOK 8 reframes the practitioner's toolkit into three precisely defined categories: Models, Methods, and Artifacts. Models are conceptual frameworks for thinking — they describe how people, communications, motivation, and change behave. Methods are documented techniques for doing — they specify how to estimate, plan, prioritize, or decide. Artifacts are tangible documents, registers, plans, and outputs produced and consumed during the project lifecycle.

This shift matters for the PMP exam because the test now expects you to recognize each item by category and to know when it is appropriate to apply it. A question may describe a leader struggling with a brand-new team member and ask which model best diagnoses the situation. Another may describe a sponsor demanding a cost estimate before requirements are stable and ask which estimation method fits. A third may describe a stakeholder complaint about lost decisions and ask which artifact should have captured the agreement. In each case, fluency with the three categories is the difference between guessing and reasoning.

Across the twelve lectures in this section you will study the leadership models (Situational and Servant), communication models (Sender-Receiver, Cross-Cultural, Gulf of Execution), motivation models (Maslow and Herzberg), the principal estimating methods (Three-Point, Analogous, Parametric), and the most-tested artifacts (Project Charter, Project Management Plan, Requirements Documentation, Scope Statement, WBS, Risk Register, Issue Log, Change Log, Status Reports). Together these form the integrated PM toolkit you will rely on in every performance domain and in real project delivery.

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Introduction to Models, Methods and Artifacts

Overview

Lecture 1 establishes the foundation for the entire section by defining the three categories PMBOK 8 uses to describe the project manager's working toolkit. Models are mental frameworks: they help you think through how teams behave, how messages get distorted, how needs drive motivation, and how change is received. Methods are repeatable techniques: they tell you how to estimate work, schedule activities, allocate resources, prioritize backlog items, and evaluate options. Artifacts are the documented outputs of the work: charters, plans, registers, logs, reports.

The lecture explains why this separation matters. A model alone produces no deliverable. A method alone, applied without a framework for interpreting the situation, can generate the wrong answer. An artifact, in turn, is only as useful as the model and method used to create it. The three categories work as a system. The PMP exam tests your ability to identify which category a scenario is asking about and to recommend the right item from that category.

By the end of this lecture you should be able to define each category in your own words, give two examples of each, and explain how they collectively replace the older Tools and Techniques language. You should also understand that PMBOK 8 lists models, methods, and artifacts that apply across all performance domains, not just within a single process group, which is a significant departure from the process-centric structure of earlier editions.

Key Concepts

- Models are frameworks for thinking — they describe relationships, behaviors, and dynamics rather than producing direct deliverables.
  - Leadership models describe how a leader adapts style to follower readiness.
  - Communication models describe how messages encode, transmit, and decode.
  - Motivation models describe what drives human effort and engagement.
- Methods are documented techniques that produce a result when applied correctly.
  - Estimating methods quantify duration, cost, or effort.
  - Scheduling methods sequence and time activities.
  - Prioritization methods rank items by value, risk, or urgency.
- Artifacts are tangible documents and registers produced and updated throughout the project lifecycle.
- PMBOK 8 replaces the Tools and Techniques chapter from earlier editions and reorganizes content around the twelve performance domains.
- The same artifact may be referenced by multiple performance domains, reflecting integrated rather than siloed project management.
- Selecting the right model, method, or artifact is itself an act of tailoring — there is no universal best choice.
- On the exam, the question stem typically contains the clue: words like 'framework,' 'theory,' or 'understand' usually signal a model; 'technique,' 'estimate,' or 'calculate' signal a method; 'document,' 'register,' or 'log' signal an artifact.
- A project manager fluent in all three categories can move from diagnosis (model) to action (method) to record (artifact) without gaps.

Key Terms & Definitions

Term	Definition
Model	A conceptual framework that describes relationships, behaviors, or dynamics in a project environment.
Method	A documented, repeatable technique that produces a specific result when applied to a problem.
Artifact	A tangible document, register, plan, log, or report produced or consumed during the project.
Performance Domain	One of twelve interrelated areas of activity that define how project work delivers value in PMBOK 8.
Tools and Techniques	The legacy PMBOK 6 chapter, now replaced by the broader Models, Methods, and Artifacts grouping in PMBOK 8.
Tailoring	The deliberate selection and adaptation of models, methods, and artifacts to suit project context.

Exam Tips

✦ EXAM TIPS

- If a question describes how someone is thinking about a situation, the answer is usually a model.
- If a question asks for a calculation or quantitative result, the answer is usually a method.
- If a question asks what document to update or where to record something, the answer is an artifact.
- PMBOK 8 vocabulary is precise — do not interchange model, method, and artifact in your answer choices.

Knowledge Check

✦ KNOWLEDGE CHECK

1. A project manager wants to diagnose why a new team member is hesitant despite being eager. Which category best applies — model, method, or artifact?
2. The sponsor asks for a duration estimate using optimistic, pessimistic, and most likely values. Which category is being invoked?
3. A stakeholder complains they never see written acceptance criteria. Which category does this concern belong to?
4. Why did PMBOK 8 replace the Tools and Techniques chapter with the Models, Methods, and Artifacts structure?

Categorizing Models, Methods and Artifacts in PMBOK 8

Overview

Lecture 2 takes the three categories introduced in Lecture 1 and shows how PMBOK 8 organizes them internally. Models are grouped by purpose — communication, motivation, change, complexity, project team development, and other situational frameworks. Methods are grouped by function — estimating, scheduling and resource allocation, and data gathering or decision-making. Artifacts are grouped by lifecycle role — strategy documents, planning documents, baselines, registers and logs, contracts and agreements, and reports.

Understanding these groupings is essential because the exam will not always announce which category a question targets. A scenario about a sponsor demanding rapid prioritization may be testing your knowledge of a decision-making method like MoSCoW. A scenario about repeated misunderstandings on a global team may be testing your knowledge of cross-cultural communication models. Recognizing the grouping accelerates correct answer selection.

The lecture also reinforces that some items are 'high-yield' for the PMP exam — they appear disproportionately often because they map directly to the People, Process, and Business Environment domains tested in the ECO. Situational Leadership, Servant Leadership, Tuckman's Ladder, the communication models, Maslow, Herzberg, Three-Point Estimation, and the major artifacts (Charter, PM Plan, WBS, Risk Register) form the core that every candidate must master.

Key Concepts

- Models are organized into six groupings in PMBOK 8.
  - Communication models
  - Motivation models
  - Change models
  - Complexity models
  - Project team development models
  - Other models such as leadership and conflict frameworks
- Methods are organized into three primary groupings.
  - Estimating methods (Three-Point, Analogous, Parametric, Bottom-Up)
  - Scheduling and resource allocation methods
  - Data gathering, analysis, and decision-making methods
- Artifacts are organized into six lifecycle-based groupings.
  - Strategy artifacts such as the business case and benefits management plan
  - Logs and registers including the risk register and issue log
  - Plans including the project management plan and subsidiary plans
  - Hierarchy charts such as the WBS and OBS
  - Baselines including scope, schedule, and cost
  - Reports such as status, performance, and forecast reports
- The same model may apply in more than one performance domain — Servant Leadership informs both the Team and Stakeholder domains.
- Methods are often paired with artifacts — Three-Point Estimation produces inputs to the cost and schedule baselines.
- Recognizing the grouping in a question stem helps eliminate distractors quickly.
- High-yield items appear on most exam forms and deserve deep study.

- PMBOK 8 expects you to know not just the name of each item but its purpose, inputs, and typical outputs.

Key Terms & Definitions

Term	Definition
Communication Model	A framework that describes how messages encode, transmit, and decode between sender and receiver.
Estimating Method	A technique for quantifying duration, cost, or effort, including Three-Point, Analogous, Parametric, and Bottom-Up.
Hierarchy Chart	An artifact that decomposes work, organization, or resources into nested levels — examples include the WBS and OBS.
Baseline	An approved version of a plan against which performance is measured — scope, schedule, and cost baselines are required.
Register	An artifact used to track items of a similar type, such as risks, issues, or stakeholders.
Performance Report	An artifact that communicates project status, forecasts, and variances to stakeholders.

Exam Tips

✦ EXAM TIPS

- Memorize which grouping each high-yield item belongs to — exam stems often hint at the grouping rather than naming the item directly.
- When a question references a 'plan,' check whether it means the overall PM Plan or a subsidiary plan; they are distinct artifacts.
- Recognize that 'log' and 'register' are both artifact subtypes but track different things — issues versus risks.
- If a question lists multiple models in the options, choose the one whose grouping matches the scenario's focus area.

Knowledge Check

✦ KNOWLEDGE CHECK

5. Into which grouping would you place Tuckman's Ladder?
6. Which artifact grouping contains the Project Management Plan?
7. Name two estimating methods that fall in the methods category.

8. Why is Servant Leadership relevant across multiple performance domains?

The Situational Leadership Model

Overview

Situational Leadership, developed by Paul Hersey and Ken Blanchard, is one of the most frequently tested leadership models on the PMP exam. Its central insight is simple: there is no single best leadership style. The right style depends on the development level of the person being led. A skilled leader does not lead the whole team the same way — they read each individual and adapt.

The model defines four development levels (D1 through D4) based on competence and commitment, and four corresponding leadership styles (S1 through S4) based on directive and supportive behavior. The leader's job is to match style to level: Directing for D1, Coaching for D2, Supporting for D3, and Delegating for D4. Misalignment — for example, delegating to a new team member or directing a seasoned expert — predictably produces frustration, disengagement, or rework.

The exam loves scenario questions where you must match a described situation to the correct development level and the correct style. Mastering the four-by-four mapping, and recognizing the linguistic cues that signal each level, is the highest-value investment you can make in this lecture.

Key Concepts

- Development levels are defined by two dimensions: competence (skill) and commitment (motivation).
  - D1: low competence, high commitment — enthusiastic beginner
  - D2: some competence, low commitment — disillusioned learner
  - D3: high competence, variable commitment — capable but cautious performer
  - D4: high competence, high commitment — self-reliant achiever
- Leadership styles vary in two dimensions: directive behavior (task focus) and supportive behavior (relationship focus).
  - S1 Directing: high directive, low supportive
  - S2 Coaching: high directive, high supportive
  - S3 Supporting: low directive, high supportive
  - S4 Delegating: low directive, low supportive
- Matching style to level is the core skill — D1 with S1, D2 with S2, D3 with S3, D4 with S4.
- Development level is task-specific, not person-specific — the same person may be D4 in one domain and D1 in another.
- Mismatches produce predictable failure patterns: over-directing seasoned experts causes disengagement; under-supporting beginners causes panic.
- Situational Leadership is dynamic — as the team member grows, the leader must consciously shift style.
- The model is descriptive of how leaders should adapt, not prescriptive of personality — anyone can learn all four styles.
- On the exam, identify the development level first, then choose the style that matches.

Key Terms & Definitions

Term	Definition
Situational Leadership	A leadership model in which the leader adapts style based on the development level of the follower for a given task.
Development Level (D1-D4)	A classification of a team member's competence and commitment for a specific task.
Directing (S1)	A high-task, low-relationship style appropriate for D1 followers who need clear instructions.
Coaching (S2)	A high-task, high-relationship style for D2 followers who have lost initial confidence.
Supporting (S3)	A low-task, high-relationship style for D3 followers who have skill but need encouragement.
Delegating (S4)	A low-task, low-relationship style for D4 followers who can perform autonomously.

Exam Tips

✦ EXAM TIPS

- Read the scenario for competence cues ('experienced,' 'new to,' 'first time') and commitment cues ('eager,' 'frustrated,' 'confident').
- When the team member is new and enthusiastic, Directing is correct even though 'just tell them what to do' sounds heavy-handed.
- When the team member is competent but recently discouraged, Supporting beats Delegating — they need encouragement, not autonomy.
- Delegating is only correct when both competence and commitment are high; do not pick it just because the person is experienced.

Knowledge Check

✦ KNOWLEDGE CHECK

9. A new hire eagerly volunteers for a task they have never done. What development level and leadership style apply?
10. A senior engineer is suddenly missing deadlines after a personal setback but has the skill to deliver. What style should the PM adopt?
11. Why is Delegating the wrong choice for a D2 team member even if they were once strong performers?
12. Explain why development level is task-specific rather than a fixed personality trait.

The Servant Leadership Model

Overview

Servant Leadership inverts the traditional command-and-control hierarchy. The servant leader exists to serve the team, removing obstacles, providing resources, and creating the conditions under which others can succeed. Coined by Robert Greenleaf in 1970, the model has become foundational to agile and hybrid project leadership and is heavily weighted in PMBOK 8 and on the PMP exam.

Greenleaf identified ten characteristics: listening, empathy, healing, awareness, persuasion, conceptualization, foresight, stewardship, commitment to growth of people, and building community. These are not soft traits — they are practical behaviors that demonstrably improve team cohesion, decision quality, and delivery speed when consistently practiced.

The exam tests both the philosophy (recognizing servant leadership when described) and its contrast with transactional leadership (recognizing when a leader is failing to serve). Many scenario questions describe a dysfunctional team and ask which leadership approach would correct course; servant leadership is frequently the right answer in modern project contexts.

Key Concepts

- Servant leadership inverts the org chart — the leader serves the team rather than the team serving the leader.
- Greenleaf identified ten characteristics of servant leaders.
  - Listening, empathy, healing
  - Awareness, persuasion, conceptualization
  - Foresight, stewardship
  - Commitment to growth of people, building community
- Servant leadership is the dominant style in agile and hybrid environments, and it underpins much of PMBOK 8's people domain.
- It contrasts with transactional leadership, which is based on reward and punishment for performance.
- Practical servant-leader behaviors include removing impediments, shielding the team from disruption, asking 'how can I help?', and amplifying team accomplishments.
- Servant leadership grows team capability over time — the goal is for the team to need the leader less, not more.
- It is not weak leadership: servant leaders still hold standards, make hard calls, and confront poor performance — but they do so to serve the team's long-term success.
- Servant leadership aligns with PMBOK 8 principles of stewardship, creating a collaborative team environment, and demonstrating leadership behaviors.

Key Terms & Definitions

Term	Definition
Servant Leadership	A leadership philosophy in which the leader's primary role is to serve and develop the team.
Stewardship	Acting as a trusted caretaker of the team, organization, and resources entrusted to the leader.

Term	Definition
Persuasion	Influencing through reasoned dialogue rather than positional authority.
Foresight	The ability to anticipate consequences and prepare for likely future events.
Empathy	Understanding and acknowledging the perspectives and feelings of others.
Transactional Leadership	A leadership style based on exchange — rewards for performance, sanctions for failure — contrasted with servant leadership.

Exam Tips

✦ EXAM TIPS

- If a scenario describes a leader removing impediments and amplifying team success, the answer is almost always servant leadership.
- Beware of distractors that describe servant leadership as 'passive' — it is active and demanding, not passive.
- Servant leadership and agile project management often appear together; recognize the pairing.
- If asked what the project manager should do first in a dysfunctional agile team, the servant-leadership answer is usually to ask the team what is blocking them.

Knowledge Check

✦ KNOWLEDGE CHECK

13. Name four of Greenleaf's ten characteristics of servant leaders.
14. How does servant leadership differ from transactional leadership?
15. Why is servant leadership especially well suited to agile and hybrid environments?
16. A team is missing sprint goals because of unresolved cross-team dependencies. What servant-leader action addresses this?

Communication Models in Project Management

Overview

Communication is the project manager's single largest time investment, and PMBOK 8 introduces multiple models to help diagnose breakdowns. The Sender-Receiver model is the foundational frame: a sender encodes a message, transmits it through a medium, and the receiver decodes it. Noise — physical, semantic, or psychological — can distort the signal at any step. Feedback closes the loop and confirms understanding.

PMBOK 8 also introduces cross-cultural communication models that account for differences in language, idiom, gesture, hierarchy expectations, and time orientation. These models matter for global teams and become high-yield on the exam as workforces become more distributed. A third class — the Gulf of Execution and Evaluation — explains how users (or stakeholders) experience the gap between intent and outcome and is increasingly cited in PMBOK 8 product-development contexts.

The lecture's goal is to make you fluent in diagnosing where a communication breakdown occurred (encoding, transmission, noise, decoding, feedback) so that you can recommend the appropriate corrective action — paraphrasing, choosing a different medium, slowing down, adapting tone, or adding a feedback loop.

Key Concepts

- The Sender-Receiver model has five elements.
  - Sender — the originator of the message
  - Encoding — translating thought into language or symbols
  - Transmission medium — email, chat, voice, video, in-person
  - Decoding — the receiver's interpretation
  - Feedback — confirmation of received meaning
- Noise can be physical (a noisy room), semantic (jargon, ambiguity), or psychological (bias, distraction).
- Feedback is what distinguishes effective communication from broadcasting — without it, the loop is open.
- Cross-cultural communication models address differences across.
  - High-context vs low-context cultures
  - Direct vs indirect communication styles
  - Hierarchical vs egalitarian expectations
  - Linear vs flexible time orientation
- The Gulf of Execution describes the gap between user intent and the system's required action; the Gulf of Evaluation describes the gap between system state and user understanding.
- Channel selection matters — rich media (video, in-person) is appropriate for ambiguous or sensitive messages; lean media (email, chat) for simple confirmations.
- Active listening, paraphrasing, and explicit confirmation reduce decoding errors.
- Communication breakdowns are the root cause of most project conflicts cited on the PMP exam.

Key Terms & Definitions

Term	Definition
Sender-Receiver Model	A communication model in which a sender encodes and transmits a message, the receiver decodes it, and feedback closes the loop.
Noise	Anything that distorts a message between sender and receiver — physical, semantic, or psychological.
Feedback Loop	The receiver's confirmation back to the sender that the message was correctly understood.
High-Context Culture	A culture in which meaning is conveyed more through context, relationship, and implication than explicit words.
Gulf of Execution	The gap between what a user intends and the actions a system requires to fulfill that intent.
Active Listening	A communication technique involving full attention, paraphrasing, and clarifying questions to confirm understanding.

Exam Tips

✦ EXAM TIPS

- If a scenario describes the same message being misunderstood by different people, look for semantic noise or cross-cultural mismatch.
- When the right answer involves 'confirming understanding,' the underlying concept is feedback in the Sender-Receiver model.
- For sensitive or complex messages, choose richer media (video, in-person) over email.
- Cross-cultural questions often have a distractor that suggests imposing one culture's style on the other — that is rarely correct.

Knowledge Check

✦ KNOWLEDGE CHECK

17. Name the five elements of the Sender-Receiver communication model.
18. Distinguish physical, semantic, and psychological noise with one example each.
19. Why is video preferable to email for delivering bad news to a key stakeholder?
20. How does a high-context culture differ from a low-context one in communication style?

Maslow's Hierarchy of Needs

Overview

Maslow's Hierarchy of Needs is the most widely recognized motivation model in management literature and a reliable exam topic. Abraham Maslow proposed that humans are motivated by a sequence of needs, organized in a pyramid from most basic (physiological) to most aspirational (self-actualization). Lower needs generally dominate motivation until they are reasonably satisfied; higher needs then come to the fore.

For project managers, the model offers a diagnostic lens. A team member who skips meals, sleeps too little, or fears layoff is operating in the lower tiers and will not be motivated by recognition or growth opportunities until those basics are addressed. Conversely, a team member whose basic needs are met may be disengaged precisely because the work does not offer belonging, esteem, or self-actualization.

While Maslow's model has been critiqued for its rigidity and cultural assumptions, the PMP exam tests the classic five-level pyramid and expects you to map motivational interventions to the correct tier. Recognizing the tier of a described need is the central exam skill.

Key Concepts

- Maslow's pyramid has five levels, from bottom to top.
  - Physiological — food, water, sleep, shelter
  - Safety — physical security, job security, health, financial stability
  - Belonging — friendships, team membership, family ties
  - Esteem — recognition, respect, achievement, status
  - Self-actualization — fulfilling one's potential, creativity, growth
- Lower needs generally must be reasonably satisfied before higher needs strongly motivate behavior.
- Motivators that target the wrong tier fail — bonus pay rarely motivates someone whose primary unmet need is belonging.
- Project context affects which tier is most salient — a layoff rumor activates safety needs across the whole team.
- Recognition programs target the esteem tier and can be powerful when basic and belonging needs are met.
- Self-actualization-tier motivators include stretch assignments, autonomy, mastery opportunities, and meaningful work.
- Maslow's model complements Herzberg's Two-Factor theory — both will appear in this section.
- On the exam, identify the unmet need described in the scenario, then choose the intervention that addresses that tier.

Key Terms & Definitions

Term	Definition
Hierarchy of Needs	Maslow's five-tier model of human motivation, from physiological needs at the base to self-actualization at the top.
Physiological Needs	Basic biological requirements such as food, water, sleep, and shelter.

Term	Definition
Safety Needs	Need for security, stability, and protection from harm — including job security in a work context.
Belonging Needs	Need for social connection, team membership, and acceptance.
Esteem Needs	Need for recognition, respect, achievement, and status.
Self-Actualization	The need to fulfill one's potential through creativity, mastery, and meaningful work.

Exam Tips

✦ EXAM TIPS

- Match the intervention to the tier of the unmet need; cash bonuses rarely satisfy belonging or self-actualization needs.
- If a scenario describes layoff rumors or unstable funding, the relevant tier is safety, not esteem.
- Recognition is an esteem-tier lever — useful only when belonging and lower tiers are already satisfied.
- Avoid distractors that propose a single universal motivator; Maslow's premise is that motivation is tier-dependent.

Knowledge Check

✦ KNOWLEDGE CHECK

21. List Maslow's five tiers in order from base to apex.
22. A team member is anxious about an impending reorganization. Which tier is activated?
23. Why might a bonus fail to motivate a disengaged senior contributor?
24. Give an example of a self-actualization-tier motivator a PM can offer.

Herzberg's Two-Factor Theory

Overview

Frederick Herzberg's Two-Factor Theory, sometimes called the Motivation-Hygiene Theory, complements Maslow by separating workplace factors into two distinct groups. Hygiene factors — salary, working conditions, company policy, supervision, job security — do not motivate when present; they only demotivate when absent. Motivators — achievement, recognition, the work itself, responsibility, advancement, growth — generate genuine engagement when present.

The practical implication is profound: raising salary fixes dissatisfaction but does not produce satisfaction. To create real motivation, project managers must invest in the motivator side — giving people meaningful work, visible achievement, and growth opportunities. This explains why some highly-paid teams remain disengaged and some modestly-paid teams thrive.

On the PMP exam, Herzberg appears in scenarios where a team is paid well, has good benefits, and yet morale is poor. The correct answer typically addresses motivators — recognition, expanded responsibility, growth conversations — rather than additional hygiene-tier investments.

Key Concepts

- Herzberg distinguishes hygiene factors from motivators.
  - Hygiene factors prevent dissatisfaction but do not create satisfaction
  - Motivators create genuine satisfaction and engagement
  - The opposite of dissatisfaction is not satisfaction; it is no dissatisfaction
- Common hygiene factors include.
  - Salary and benefits
  - Working conditions and physical environment
  - Company policies and administration
  - Quality of supervision
  - Job security
- Common motivators include.
  - Achievement and the work itself
  - Recognition for accomplishment
  - Responsibility and ownership
  - Advancement and growth
  - Learning and mastery
- Improving hygiene factors removes dissatisfaction but does not produce motivation — they are necessary but not sufficient.
- Motivators must be intrinsic to the work to be sustainable — extrinsic praise without meaningful work fades quickly.
- Herzberg's theory aligns with PMBOK 8's emphasis on creating a productive team environment and demonstrating leadership behaviors.
- The theory also explains why pay raises alone rarely solve engagement problems on high-performing teams.
- Apply the theory by auditing both factor sets: ensure hygiene factors are acceptable, then design the work to deliver motivators.

Key Terms & Definitions

Term	Definition
Two-Factor Theory	Herzberg's motivation theory distinguishing hygiene factors from motivators.
Hygiene Factors	Workplace factors whose absence creates dissatisfaction but whose presence does not motivate — salary, conditions, policies.
Motivators	Intrinsic workplace factors that produce genuine satisfaction — achievement, recognition, responsibility, growth.
Job Enrichment	Designing work to include more motivators — variety, autonomy, meaningful responsibility — to drive engagement.
Intrinsic Motivation	Motivation that arises from the work itself rather than from external rewards.
Dissatisfaction	The negative emotional state resulting from absent or inadequate hygiene factors, distinct from lack of satisfaction.

Exam Tips

✦ EXAM TIPS

- If a scenario describes a well-paid but disengaged team, the answer involves motivators — recognition, responsibility, growth — not hygiene.
- Distinguish 'no dissatisfaction' from 'satisfaction' on the exam; they are different states in Herzberg's framework.
- Hygiene improvements are necessary as a foundation; do not ignore them, but recognize their limits.
- Job enrichment is Herzberg's preferred intervention for restoring motivation.

Knowledge Check

✦ KNOWLEDGE CHECK

25. Name three hygiene factors and three motivators in Herzberg's theory.
26. Why does a salary increase often fail to improve long-term motivation?
27. Define job enrichment and give one project-management example.
28. How does Herzberg's theory complement Maslow's hierarchy?

Three-Point Estimation, Analogous and Parametric Estimation

Overview

Estimation is one of the most heavily tested method categories on the PMP exam. Three-Point Estimation, Analogous Estimation, and Parametric Estimation each have specific use cases, accuracy profiles, and required inputs. Mastering when to choose each — and how to compute the Three-Point result — is essential.

Three-Point Estimation, also called PERT (Program Evaluation and Review Technique), uses three values per activity: optimistic (O), most likely (M), and pessimistic (P). The Beta distribution formula (O + 4M + P) divided by 6 produces the expected duration. Standard deviation is (P minus O) divided by 6. Three-Point Estimation is preferred when uncertainty is meaningful and you need a probability-aware estimate.

Analogous Estimation uses historical data from similar past projects to forecast the current work — it is fast, low-cost, and lower-accuracy. Parametric Estimation uses a unit rate or formula (cost per square foot, hours per line of code) multiplied by the project's scope quantity. The correct choice depends on the maturity of historical data, the level of detail available, and the accuracy required.

Key Concepts

- Three-Point Estimation formula (Beta/PERT): Expected = (O + 4M + P) divided by 6.
- Three-Point Standard Deviation: SD = (P minus O) divided by 6.
- Triangular Three-Point alternative: Expected = (O + M + P) divided by 3, used when no probability weighting is desired.
- Analogous Estimation.
  - Uses one or more similar past projects as the reference
  - Fast and inexpensive — often used in early phases
  - Accuracy depends on the similarity of past projects
  - Often called Top-Down estimating
- Parametric Estimation.
  - Uses a unit rate or statistical relationship: rate times quantity
  - Requires reliable historical or industry parameters
  - Scales linearly with quantity
  - Examples include cost per square foot, hours per function point
- Bottom-Up Estimation aggregates estimates from the lowest WBS level upward — most accurate, most time-consuming.
- Accuracy ranking, generally: Bottom-Up, then Three-Point, then Parametric, then Analogous.
- Cost of estimating ranking, generally: Bottom-Up highest, then Three-Point, then Parametric, then Analogous — inverse to accuracy.
- Choose method based on data availability, required accuracy, and phase of the project.

Key Terms & Definitions

Term	Definition
Three-Point Estimation	A technique that uses optimistic, most likely, and pessimistic values per activity to compute an expected duration or cost.
PERT	Program Evaluation and Review Technique — the beta-weighted Three-Point formula (O + 4M + P) divided by 6.
Analogous Estimation	An estimating method that uses historical data from similar past projects to forecast the current effort.
Parametric Estimation	An estimating method that multiplies a unit rate by a quantity, based on a statistical or historical relationship.
Bottom-Up Estimation	Aggregating detailed estimates from the lowest WBS elements upward to produce a total project estimate.
Standard Deviation (PERT)	A measure of estimating uncertainty computed as (P minus O) divided by 6 in Three-Point Estimation.

Exam Tips

✦ EXAM TIPS

- Memorize the PERT formula: (O + 4M + P) divided by 6 — it is a near-certainty on the exam.
- Standard Deviation is (P minus O) divided by 6; Variance is the square of SD.
- Analogous is fast and cheap but inaccurate — use it early or when data is scarce.
- Parametric requires a reliable rate — verify the rate before applying it to a new project.

Knowledge Check

✦ KNOWLEDGE CHECK

29. Compute the PERT expected duration when O = 4, M = 8, P = 18.
30. Compute the PERT standard deviation for the same values.
31. When is Analogous Estimation preferred over Parametric Estimation?
32. Rank the four estimation methods by typical accuracy.

Key Artifacts: Project Charter and Project Management Plan

Overview

The Project Charter and the Project Management Plan are the two most foundational artifacts in any project. The Charter formally authorizes the project, names the project manager, and establishes high-level scope, objectives, success criteria, and key stakeholders. It is issued by the sponsor or initiator and gives the project manager the authority to apply organizational resources. Without it, there is no project.

The Project Management Plan is the comprehensive document that integrates all subsidiary plans — scope, schedule, cost, quality, resources, communications, risk, procurement, and stakeholder engagement — along with baselines and configuration management approach. It is the master reference for executing, monitoring, and controlling the project, and it is updated only through approved change requests once baselined.

On the exam, expect questions about what each artifact contains, who creates and approves it, when it is created in the lifecycle, and how it relates to subsidiary plans and baselines. Distinguishing Charter content from PM Plan content is a frequent test point.

Key Concepts

- The Project Charter formally authorizes the project and grants the PM authority to act.
- The Charter typically includes.
  - Project purpose and high-level scope
  - Measurable project objectives and success criteria
  - High-level requirements, risks, and assumptions
  - Summary milestone schedule and budget
  - Stakeholder list and PM assignment with authority level
  - Sponsor name and approval
- The Charter is issued by the sponsor or initiator — not authored by the PM in isolation.
- The Project Management Plan integrates all subsidiary plans and baselines.
  - Scope, schedule, cost, and quality management plans
  - Resource, communications, risk, procurement, and stakeholder management plans
  - Scope, schedule, and cost baselines
  - Change management and configuration management approach
- The PM Plan is progressively elaborated — sections are added and refined as the project advances.
- Once baselined, the PM Plan can only be changed through approved change requests via integrated change control.
- The PM Plan is the primary input to executing, monitoring, and controlling processes.
- The Charter authorizes; the PM Plan executes. Confusing them on the exam is a common error.

Key Terms & Definitions

Term	Definition
Project Charter	The artifact that formally authorizes the project and grants the project manager authority to apply resources.

Term	Definition
Project Management Plan	The integrated document containing all subsidiary plans and baselines that guide project execution.
Baseline	An approved version of scope, schedule, or cost used to measure performance and authorize changes.
Subsidiary Plan	A component of the PM Plan addressing a single knowledge area, such as the scope or risk management plan.
Sponsor	The person or group that provides resources and support and is accountable for the project's success.
Integrated Change Control	The formal process by which changes to the PM Plan or baselines are reviewed and approved or rejected.

Exam Tips

✦ EXAM TIPS

- If a question asks who authorizes the project, the answer involves the Charter and the sponsor.
- Distinguish 'authorize' (Charter) from 'plan' (PM Plan) — both are not the same.
- Baselines reside in the PM Plan; the Charter contains only high-level estimates, not baselines.
- Changes to the PM Plan after baselining always go through integrated change control.

Knowledge Check

✦ KNOWLEDGE CHECK

33. Name four items typically contained in the Project Charter.
34. Who issues the Project Charter, and why is this important?
35. List four subsidiary plans contained in the Project Management Plan.
36. How are changes to a baselined PM Plan handled?

Key Artifacts: Requirements Documentation, Scope Statement and WBS

Overview

Requirements Documentation, the Project Scope Statement, and the Work Breakdown Structure (WBS) form the scope-definition trio. Requirements Documentation captures stakeholder needs in detail — business, functional, non-functional, transition, and quality requirements. The Scope Statement translates those requirements into the project's deliverables, acceptance criteria, exclusions, assumptions, and constraints. The WBS decomposes the scope into a hierarchy of manageable work packages.

These three artifacts work as a sequence. Without clear requirements, the Scope Statement will be ambiguous. Without a sound Scope Statement, the WBS will be incomplete or misaligned. Without an accurate WBS, every downstream estimate, schedule, and assignment is at risk. The PMP exam frequently presents scope-related problems and expects you to identify which artifact is missing or defective.

Each artifact has a defined audience and purpose. Requirements are for traceability and acceptance. The Scope Statement is the agreement of what is in and out. The WBS is the structured map for assigning, estimating, and tracking work. Understanding these distinct roles prevents confusion and supports correct answers under exam time pressure.

Key Concepts

- Requirements Documentation captures stakeholder needs across multiple categories.
  - Business requirements — why the project exists
  - Stakeholder requirements — what specific stakeholders need
  - Solution requirements — functional and non-functional
  - Transition requirements — go-live and handover needs
  - Quality requirements — acceptable performance levels
- The Project Scope Statement contains.
  - Product scope description and deliverables
  - Acceptance criteria for each deliverable
  - Project exclusions — what is explicitly not included
  - Constraints and assumptions
- The Work Breakdown Structure decomposes the total scope into work packages.
  - Deliverable-oriented hierarchical decomposition
  - Includes 100% of project work — and only project work
  - Lowest level is the work package
  - Supports estimating, scheduling, and assignment
- The 100% Rule states that the WBS must include all work and nothing more.
- The Requirements Traceability Matrix links requirements to deliverables and tests for verifiable acceptance.
- The WBS Dictionary describes each WBS element in detail — work description, responsible party, acceptance criteria.
- Work packages should be small enough to estimate and assign but large enough to avoid micromanagement — typically eight to eighty hours of work.
- These three artifacts are foundational inputs to schedule, cost, and quality planning.

Key Terms & Definitions

Term	Definition
Requirements Documentation	An artifact capturing stakeholder needs across business, stakeholder, solution, transition, and quality categories.
Project Scope Statement	An artifact defining deliverables, acceptance criteria, exclusions, assumptions, and constraints.
Work Breakdown Structure (WBS)	A deliverable-oriented hierarchical decomposition of the total project scope into work packages.
Work Package	The lowest level of the WBS, used as the basis for estimating, scheduling, and assignment.
100% Rule	The principle that the WBS must include all work required to deliver the project and nothing more.
WBS Dictionary	A companion artifact describing each WBS element in detail — work content, owner, milestones, and acceptance criteria.

Exam Tips

✦ EXAM TIPS

- The WBS is deliverable-oriented, not task-oriented — distractors that describe a list of activities are usually wrong.
- Acceptance criteria belong in the Scope Statement (and traceability matrix), not in the WBS itself.
- The Requirements Traceability Matrix ensures every requirement maps to a deliverable and a verification.
- If a scenario describes uncontrolled growth in deliverables, the missing or weak artifact is usually the Scope Statement.

Knowledge Check

✦ KNOWLEDGE CHECK

37. Name the five categories of requirements typically captured in the Requirements Documentation.
38. What four content elements must the Project Scope Statement contain?
39. Define the 100% Rule for a WBS.
40. What is the lowest level of the WBS called, and what is its purpose?

Key Artifacts: Risk Register, Issue Log, Change Log and Status Reports

Overview

Risk Register, Issue Log, Change Log, and Status Reports are the four most actively used artifacts in day-to-day project execution. They are living documents — updated continuously, reviewed in team meetings, and shared with stakeholders. Mastering their distinct purposes and contents is critical because exam questions frequently turn on knowing which artifact captures what.

The Risk Register catalogs identified risks, their probability and impact, owners, and response strategies. The Issue Log captures problems that have already occurred and need resolution — distinct from risks, which are still potential. The Change Log records all change requests, their disposition, and the resulting actions. Status Reports communicate progress, variances, and forecasts to stakeholders.

A common exam trap is confusing risks (potential, in the Risk Register) with issues (actual, in the Issue Log). Another is asking where change requests are tracked — always the Change Log, not the Risk Register. Practice distinguishing these artifacts confidently.

Key Concepts

- The Risk Register captures all identified risks with the following data.
  - Risk description and category
  - Probability and impact scores
  - Risk score or priority
  - Risk owner
  - Planned response (avoid, transfer, mitigate, accept; or escalate)
  - Trigger conditions and current status
- The Issue Log captures problems that have already occurred.
  - Issue description and date raised
  - Owner and priority
  - Resolution and date closed
- Risks are potential; issues are actual — the distinction governs which artifact captures them.
- The Change Log records every change request and its disposition.
  - Change request ID and description
  - Submitter and date
  - Impact analysis summary
  - Disposition (approved, deferred, rejected) and date
  - Resulting actions and follow-up
- Status Reports communicate progress to stakeholders.
  - Schedule, cost, and scope performance
  - Variances and forecasts
  - Risks, issues, and changes since the last report
  - Upcoming milestones and decisions needed
- These artifacts are updated continuously — they are not one-time deliverables.
- Each artifact has a clear owner and audience; sharing them appropriately supports transparency and trust.

- On the exam, match the described event to the correct artifact: predicted problem to Risk Register, present problem to Issue Log, scope or plan modification to Change Log, periodic communication to Status Report.

Key Terms & Definitions

Term	Definition
Risk Register	An artifact listing all identified risks with probability, impact, owner, response strategy, and status.
Issue Log	An artifact tracking actual problems that have occurred, their owners, priorities, and resolutions.
Change Log	An artifact recording all change requests with submitter, impact analysis, disposition, and resulting action.
Status Report	A periodic artifact communicating progress, variances, forecasts, and decisions needed to stakeholders.
Risk Owner	The named individual responsible for monitoring and executing the response to a specific risk.
Risk Response Strategy	The chosen approach for a risk — avoid, transfer, mitigate, accept, or escalate for negative risks.

Exam Tips

✦ EXAM TIPS

- If the event is a present problem, the artifact is the Issue Log, not the Risk Register.
- Change requests are tracked in the Change Log; they are also referenced in the integrated change control process.
- Risk responses include 'escalate' in PMBOK 8 — for risks outside the PM's authority.
- Status Reports communicate performance — they do not replace the formal Risk Register or Issue Log.

Knowledge Check

✦ KNOWLEDGE CHECK

41. Distinguish a risk from an issue and name the artifact that captures each.
42. Name four data elements in a typical Risk Register entry.
43. Where are change requests tracked after they are submitted?
44. List four sections typically found in a Status Report.

Section 7 Summary: Building Your Complete PM Toolkit

Overview

This summary lecture consolidates the eleven preceding lectures into a working PM toolkit. It revisits the model, method, and artifact distinction, reviews the high-yield items in each category, and offers a decision framework for selecting the right tool in real situations. The goal is integration: not a list of items to memorize but a system of choices to deploy.

The lecture emphasizes that real projects rarely use a single model, method, or artifact in isolation. A typical week may involve diagnosing a team member's development level (Situational Leadership), choosing how to deliver a sensitive message (communication model), estimating a change request (Three-Point), recording the decision (Change Log), and updating stakeholders (Status Report). Fluent toggling among models, methods, and artifacts is the mark of an effective project manager.

For exam preparation, the lecture provides a study strategy: master the high-yield items first, learn the distinguishing cues that signal each category, practice scenario questions that require category identification, and build a personal cheat sheet that maps common exam stems to the correct answer category.

Key Concepts

- Master the high-yield models first.
  - Situational Leadership
  - Servant Leadership
  - Sender-Receiver and Cross-Cultural communication models
  - Maslow's Hierarchy of Needs
  - Herzberg's Two-Factor Theory
  - Tuckman's Ladder for team development
- Master the high-yield methods.
  - Three-Point Estimation with PERT formula
  - Analogous and Parametric Estimation
  - MoSCoW and other prioritization methods
  - Critical Path Method and Critical Chain
- Master the high-yield artifacts.
  - Project Charter and Project Management Plan
  - Requirements Documentation, Scope Statement, and WBS
  - Risk Register, Issue Log, Change Log
  - Status Reports and performance reports
- Use a category-first reading approach on exam questions: identify whether the stem points to a model, method, or artifact before evaluating options.
- Recognize linguistic cues — 'framework,' 'theory,' 'understand,' 'diagnose' point to models.
- Cues like 'estimate,' 'calculate,' 'compute,' 'prioritize' point to methods.
- Cues like 'document,' 'register,' 'log,' 'plan,' 'report' point to artifacts.
- Build a personal cheat sheet that maps the most common exam stems to the correct category and item.

Key Terms & Definitions

Term	Definition
Integrated PM Toolkit	The complete set of models, methods, and artifacts a project manager fluently selects from based on context.
High-Yield Item	A model, method, or artifact that appears disproportionately often on PMP exam forms and warrants deep mastery.
Category-First Reading	An exam strategy of first identifying which category (model, method, artifact) the stem targets before evaluating options.
Linguistic Cue	A keyword in an exam stem (e.g., 'framework,' 'calculate,' 'register') that signals the target category.
Personal Cheat Sheet	A study artifact in which the candidate maps common exam stems to category and item for rapid recall.
Toolkit Tailoring	Selecting and adapting models, methods, and artifacts to fit the unique project context.

Exam Tips

✦ EXAM TIPS

- Build your category-first habit during practice tests so it is automatic on exam day.
- Memorize linguistic cues — they cut answer time and reduce careless mistakes.
- Prioritize study time on high-yield items; do not let exotic models displace foundational ones.
- Practice scenario integration questions — they often weave a model, method, and artifact into one stem.

Knowledge Check

✦ KNOWLEDGE CHECK

45. Name three high-yield models you must master for the PMP exam.
46. Name three high-yield methods and one situation each is best suited for.
47. Name three high-yield artifacts and the lifecycle stage where each is most actively used.
48. Describe the category-first reading approach in one sentence.

Appendix

A. Models vs Methods vs Artifacts

Use this table for rapid category disambiguation when reading exam questions.

Category	What It Is	High-Yield Examples
Models	Frameworks for thinking and diagnosing — describe relationships, behaviors, and dynamics	Situational Leadership, Servant Leadership, Maslow, Herzberg, Sender-Receiver, Tuckman's Ladder
Methods	Documented techniques for doing — produce a quantitative or decision-oriented result	Three-Point Estimation, Analogous, Parametric, Bottom-Up, MoSCoW, Critical Path Method
Artifacts	Tangible documents and registers — produced and consumed during the project lifecycle	Project Charter, PM Plan, WBS, Risk Register, Issue Log, Change Log, Status Reports

B. Most Commonly Tested Items

These items appear most often on PMP exam forms. Prioritize them in your study plan.

Item	Category	Key Point to Remember
Situational Leadership	Model	Match development level (D1-D4) to leadership style (S1-S4)
Servant Leadership	Model	Greenleaf's ten characteristics; agile and hybrid environments
Sender-Receiver Model	Model	Encode, transmit, decode; noise types; feedback closes the loop
Maslow's Hierarchy	Model	Five tiers; lower must be satisfied before higher motivates
Herzberg's Two-Factor	Model	Hygiene factors prevent dissatisfaction; motivators create satisfaction
Three-Point Estimation (PERT)	Method	Expected = (O + 4M + P) divided by 6; SD = (P minus O) divided by 6
Analogous Estimation	Method	Historical data from similar projects; fast, low accuracy
Parametric Estimation	Method	Unit rate times quantity; requires reliable parameter
Project Charter	Artifact	Authorizes project; issued by sponsor; names PM
Project Management Plan	Artifact	Integrates subsidiary plans and baselines; changes via integrated change control
WBS	Artifact	Deliverable-oriented decomposition; obey the 100% Rule
Risk Register	Artifact	Potential events; Issue Log captures actual problems

C. Exam Cheat Sheet — Stem to Answer Mapping

When a question stem matches a left-column cue, the right column suggests the most likely correct category, model, method, or artifact.

Question Stem Cue	Likely Correct Category / Item
Stem contains 'framework,' 'theory,' 'understand,' 'diagnose'	Model
Stem contains 'estimate,' 'calculate,' 'compute,' 'prioritize'	Method
Stem contains 'document,' 'register,' 'log,' 'plan,' 'report'	Artifact
New, eager team member	Situational Leadership: D1 with Directing (S1)
Discouraged but skilled team member	Situational Leadership: D3 with Supporting (S3)
Disengaged but well-paid team	Herzberg motivators — recognition, responsibility, growth
Layoff rumor causing anxiety	Maslow safety tier — provide security and information
Sponsor wants quick budget number	Analogous Estimation
Known unit rate per square foot	Parametric Estimation
Optimistic, most likely, pessimistic values given	Three-Point Estimation (PERT)
Project not yet authorized	Project Charter required
Risk already occurred	Issue Log (not Risk Register)
Approved or rejected change requests	Change Log
Periodic stakeholder communication of progress	Status Report